

Challenge #1: Building a Marble Rolling Track

Group: Raúl, Gianna

Task description: Using the Unity game engine, create a road for marble races. The only game objects you can use are the 3D primitives and their physical components that enable the marbles' dynamics during the race. The roads should have varying degrees of incline, with forks, traps, and obstacles. At the end of the road, there should be a Roman Arch marking the finish line.

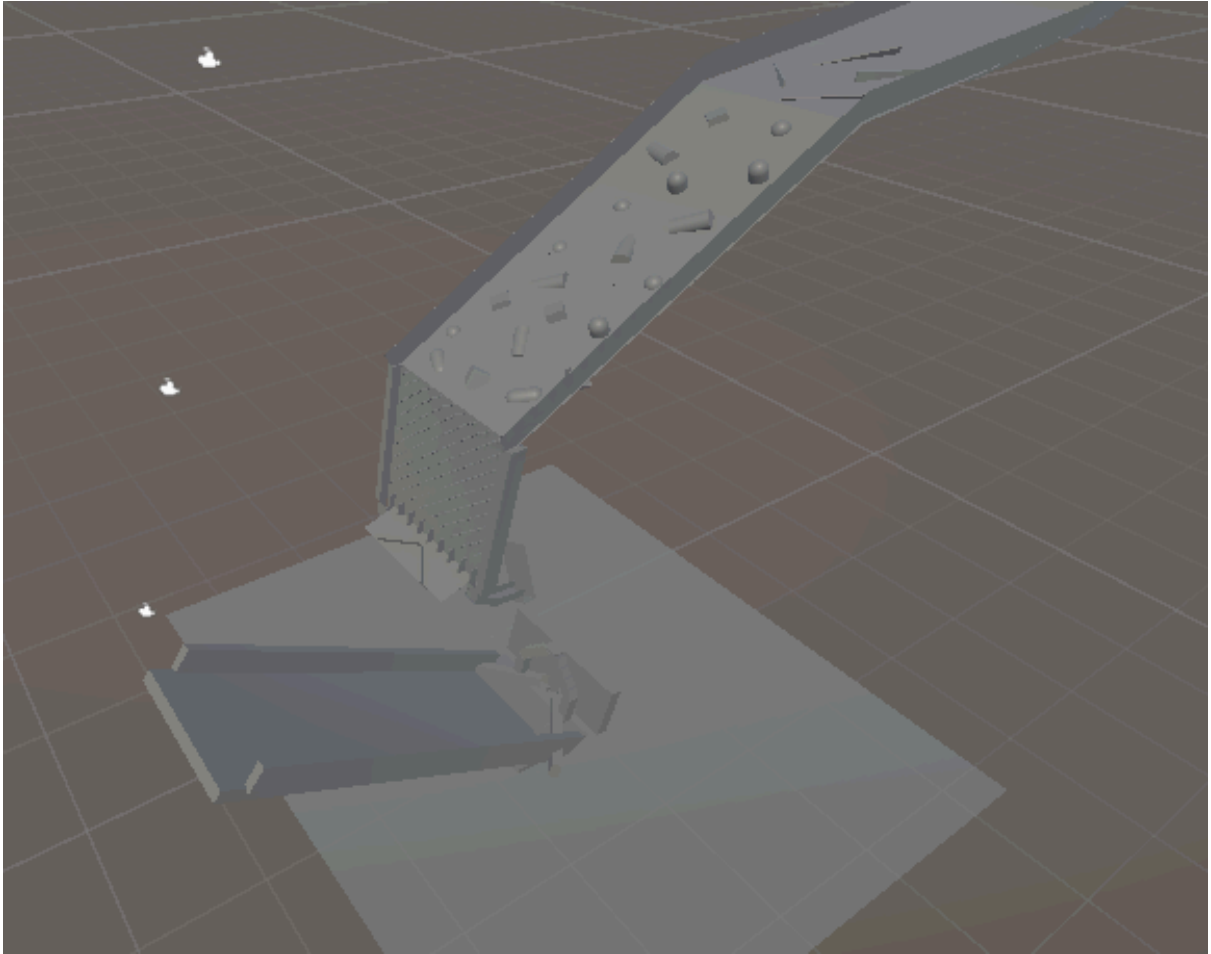
The first task was to decide what obstacles would compose the track, during this step it was decided to have a Bumpy Road, that the marbles would bump into as they went down, a Plinko Board, that the marbles would descend with a lucky few skipping the next obstacle, with the final obstacle being a Spinning Funnel, that the marbles would fall into and spin around till reaching the bottom. Then at the end of the track there would be the Roman Arch with a small bucket in which the victorious marble would rest inside, this was then made into a rough sketch of how the track would look, and then it was time to make it in Unity.

The challenge also makes use of two prefabs one for the plinko board and another for the roman arch, it also uses four material colours for the marbles, these files are included in the GitHub repository with the Racetrack scene.

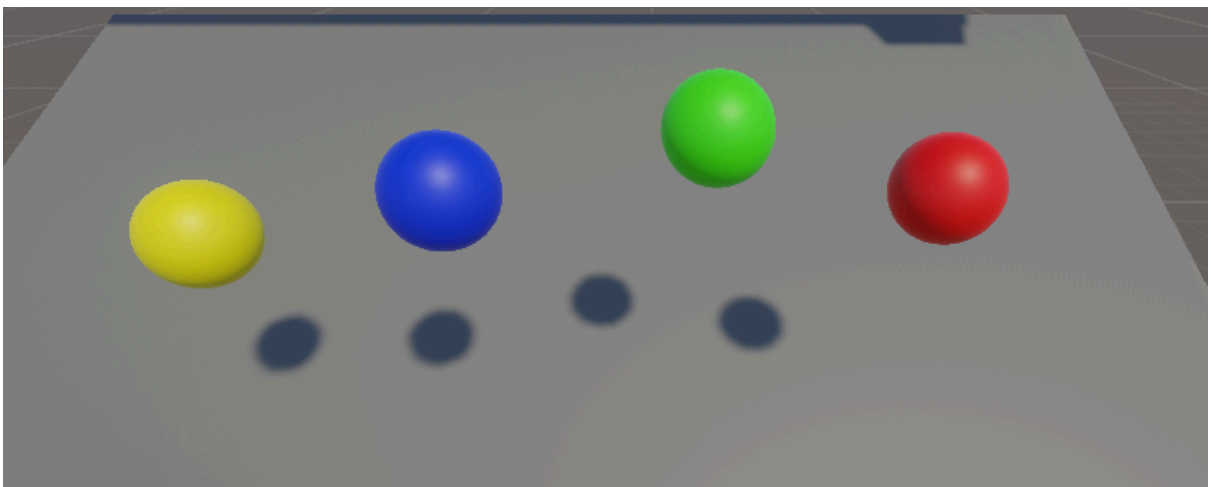
GitHub Repository Link:

<https://github.com/RaulCruzRamirez/CCOM4302-2026.git>

The Full Race:



(View of the full racetrack)



(View of the Marble Racers, each one vying for the victory cup)

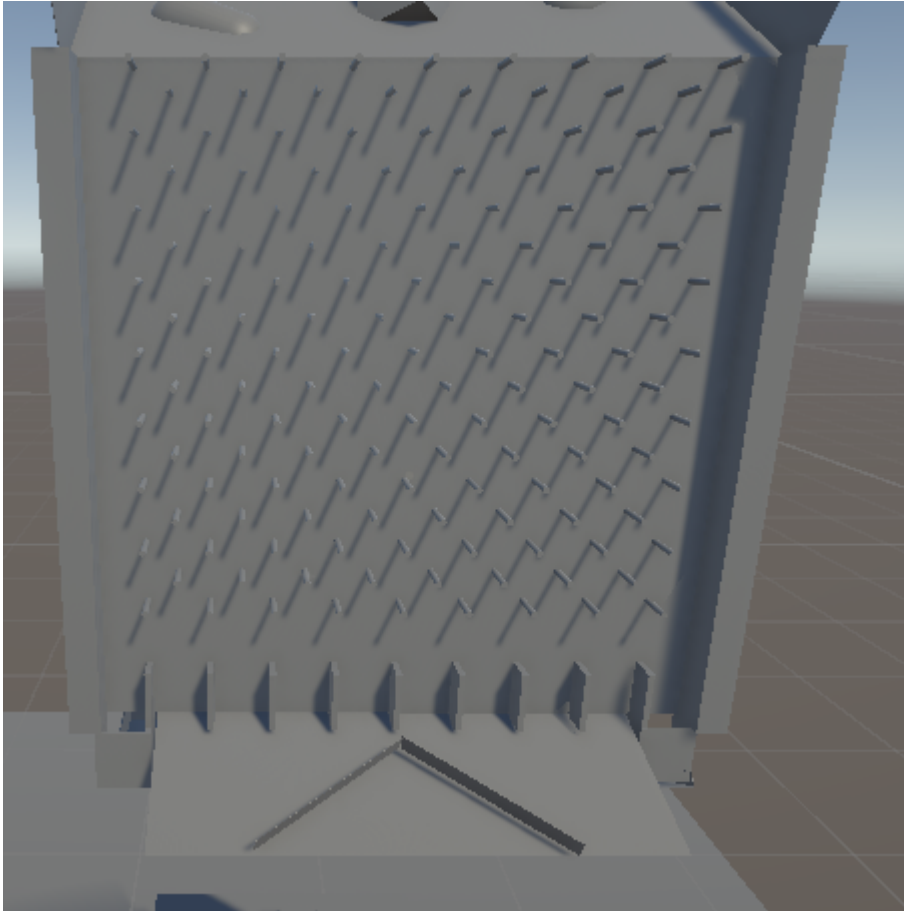
Track #1: Bumpy Road



(Top down view of the Bumpy Road)

The bumpy road is a very simple obstacle, it being a ramp sloping down with guardrails to prevent the marble from falling off the track, the obstacles come in the forms of 3d objects strewn about the ramp. These include cubes, cylinders and spheres with them being placed at random intervals along the track with different rotations, this is for the marbles to crash into and spin around. This obstacle then leads to the plinko board.

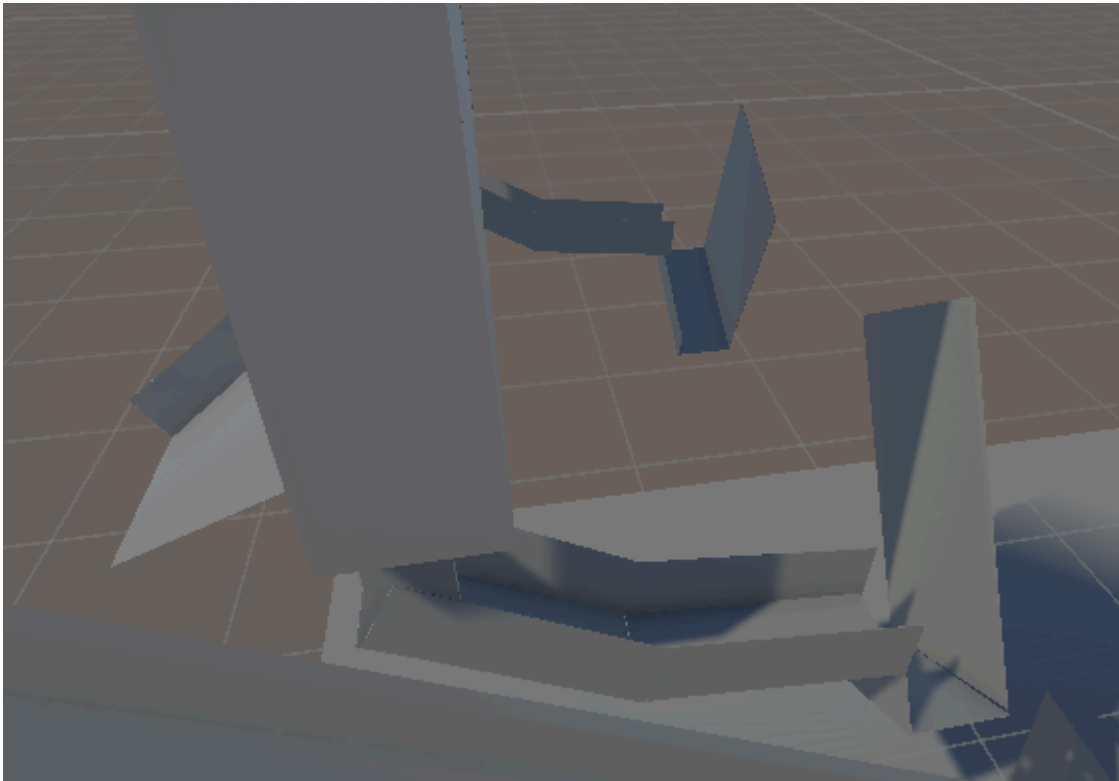
Track #2: Plinko Board



(The Plinko Board from the front)

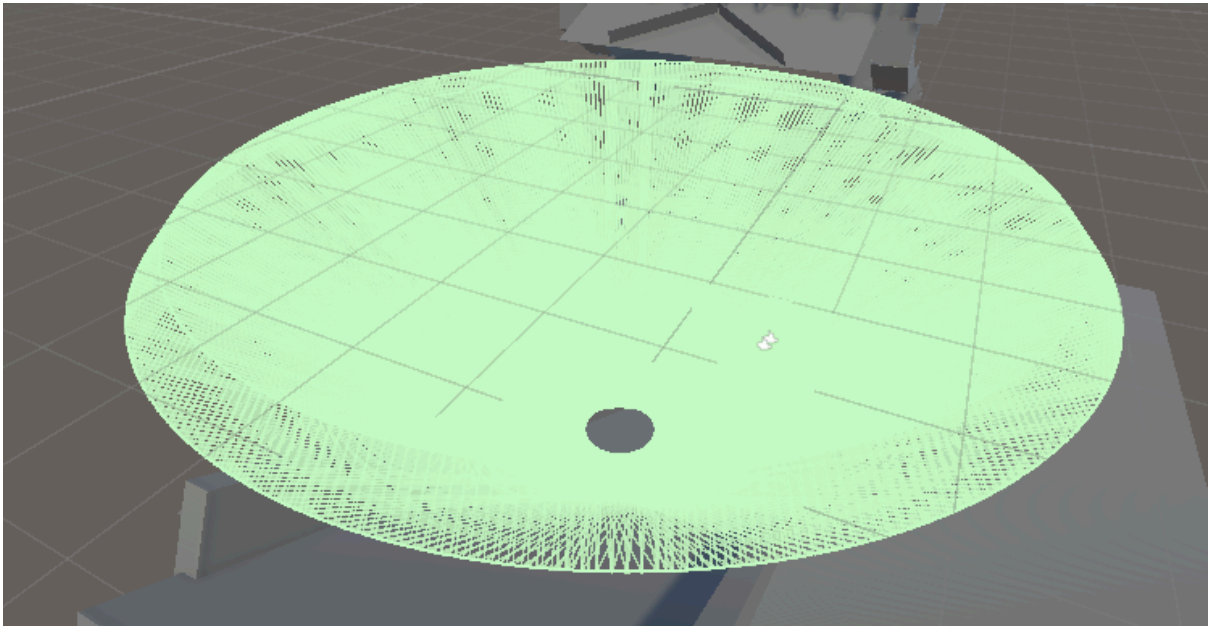
The plinko board is an obstacle in which the marble enters a vertical board and hits pegs on the way down causing the marble to move position within the board affecting where it will exit. The construction of the plinko board is made with cubes these include, the backboard where the pegs sit, the sidewalls to keep the marbles in, finish buckets which are thin walls dividing the bottom part of the board into # open parts, and a clear “glass” to keep the marbles in place which is a cube with its mesh rendering disabled to see where the marble is going. The pegs the marbles hit when going down are made of cylinders with first row having # and the second having #, the pattern repeating till it reaches the finish buckets at the end of

the board where the marbles fall. All except two of the finish buckets, the far left and right, lead down to a small ramp that splits the marbles so they fall sideways instead of straight down into the next obstacle, with the two exceptions happening when the marble reaches the far left or right of the board causing them to fall straight down into a shortcut that lets them skip the final obstacle and go straight to the finish line.



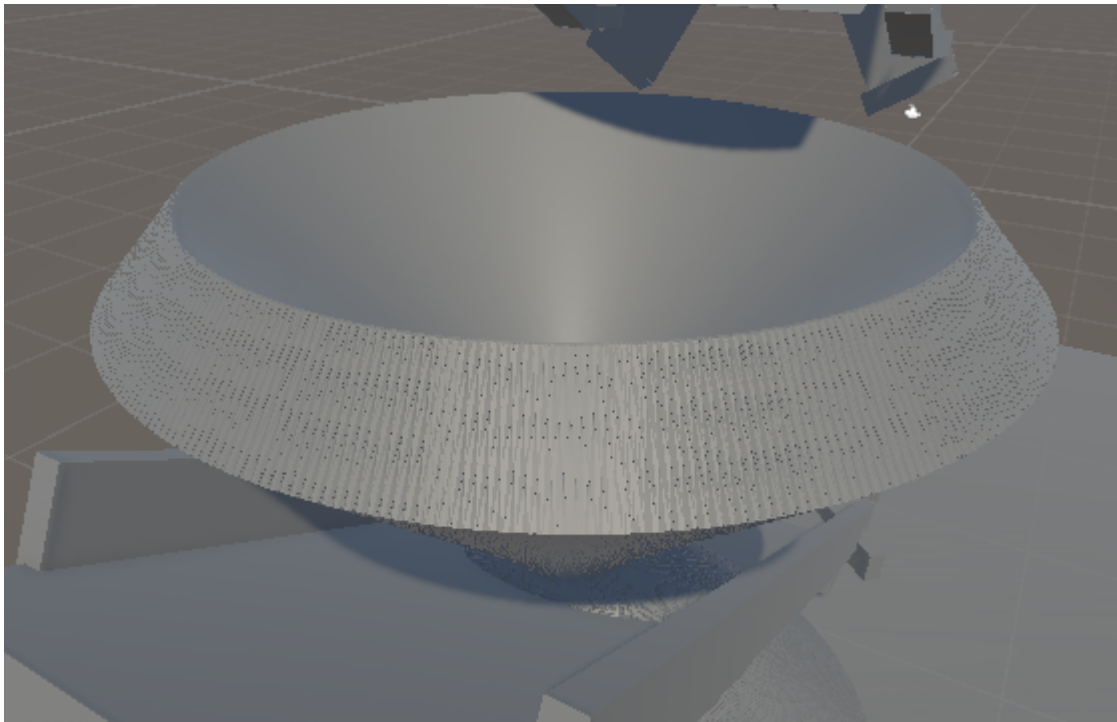
(The Plinko Board from the side)

Track #3: Spinning Funnel



(The Spinning Funnel's collision)

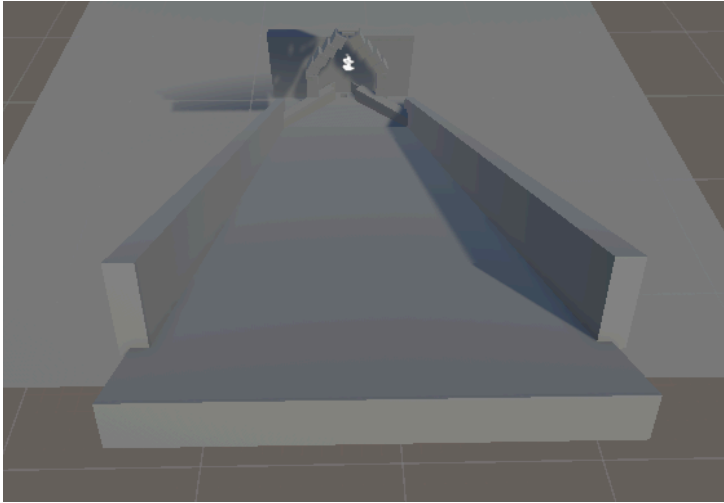
For the marbles that did not manage to get lucky and skip the final obstacle, they instead fall into a funnel that they spin around in before exiting through the bottom and onto the finish line. The funnel is made of cubes that have elongated and placed right next to each other with a rotation of 1 on the z axis, putting enough of them together so that they form a funnel that the marbles can smoothly spin around inside. Though this process creates a funnel it does not give it an exit as the collision of the cubes block out the bottom, to solve this issue the collision of all the cubes can be modified so that they spread out still forming a funnel but with enough space at the bottom for an exit. In this specific case the mesh of the blocks still stayed in place causing the funnel to appear in this form:



(The Spinning Funnel's mesh rendering)

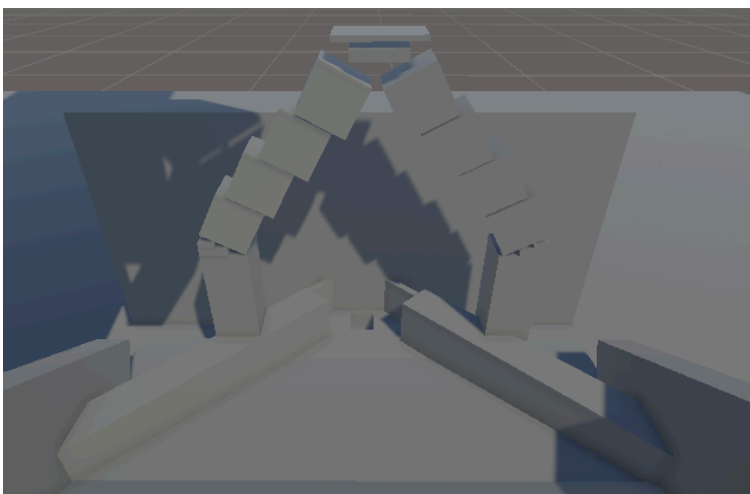
Disabling the mesh kept the funnel collision intact, though now it is invisible which can be considered an extra obstacle within itself as one would expect the marbles to fall straight into the finish line instead of spinning around.

Track #4: Finish Line



(The Finish Line from the top of the ramp)

The finish line is a simple inclined ramp with guardrails, made of cubes, that leads to the finish arch where the first marble to cross it wins the race. The marbles will enter the finish line through the funnel near the top of the ramp or through the lucky shortcut from the plinko board that lets them skip the funnel and land further down the track. At the end we find a roman arch that leads to a small cup where only one marble can fit inside, that marble being the victorious one.



(The Roman Arch at the end of the Finish Line with the small victory cup)

Experience Gained:

Raúl:

This challenge has taught me a lot about Unity, especially the areas concerning 3d Objects and how to modify them. Since in this exercise we only had access to the most basics of 3d objects one had to get creative when creating the obstacles, which amounted to an excessive amount of modifying cubes in scale and rotation. The challenge was to create a racing course and as the marbles could not propel themselves forward this made it so that the entire track was made at an incline in order to take advantage of the rigidbody property, that in turn allows the marbles to be affected by gravity allowing them to fall downhill. And another focus of the challenge was collisions and how the marbles were affected when colliding with other objects, being a main focus of the exercise it was interesting to see how this property worked in action and how for the most part it worked as expected. Though there were times when the marbles would phase through certain objects when they reached high enough speeds, this was fixed by making the objects have a bigger collision box or by putting obstacles in place in order to slow down the marble. Finally the funnel was one of the bigger problems to tackle as we only had the basic 3d objects to work with but it was overcome by using a large amount of close together cubes and rotating each one ever so slightly until they formed a funnel, then deactivating the mesh render for a surprise, we modify the collision parameters until they form a funnel with a hole at the bottom for the marbles. Doing this exercise I learned a lot about the basics of Unity, basics that have hopefully built a strong foundation for future works and more advanced techniques inside of the engine.